

SARAH C. SHI

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EDUCATION

- 2024- UNIVERSITY OF CALIFORNIA, BERKELEY, DOCTOR OF PHILOSOPHY IN EARTH SCIENCE
Supervisor: Professor Penny Wieser
- 2021-2022 UNIVERSITY OF CAMBRIDGE, MASTER OF PHILOSOPHY IN EARTH SCIENCE
Thesis: Development of Machine Learning Methodology for Automating Petrography and Studying Icelandic Xenoliths
Supervisor: Professor John MacLennan
- 2016-2020 COLUMBIA UNIVERSITY, BACHELOR OF ARTS IN EARTH SCIENCE, *summa cum laude*
Thesis: Run-Up and Syn-Eruptive Dynamics of Volcán de Fuego's Eruption of 2018
Supervisor: Professor Terry Plank

FELLOWSHIPS AND AWARDS

- 2026 BEST POSTER, GORDON CONFERENCE ON THE GEOCHEMISTRY OF MINERAL DEPOSITS
- 2026 OUTSTANDING GRADUATE STUDENT INSTRUCTOR AWARD, UNIVERSITY OF CALIFORNIA, BERKELEY
· Granted award in recognition of teaching for EPS100B (Genesis and Origin of Rocks)
- 2025 HEARTS TO HUMANITY ETERNAL (H2H8) GRADUATE RESEARCH GRANT TO ADVANCE HUMANITY
· Awarded grant (\$10,000) to examine Cascade volcanism, with a focus on 'advancing humanity'
- NORTHERN CALIFORNIA GEOLOGICAL SOCIETY, RICHARD CHAMBERS MEMORIAL SCHOLARSHIP
· Awarded scholarship (\$2,000) to examine Cascade volcanism in Northern California
- 2024-2026 CHANCELLOR'S FELLOWSHIP, UNIVERSITY OF CALIFORNIA, BERKELEY
· Awarded fellowship recognizing scholarship and academic contributions in the top admitted doctoral students applying to the Math and Physical Sciences Division; granted full tuition and fees for 3 semesters and 2 summers (\$58,000 total)
- 2024 GATES CAMBRIDGE SCHOLARSHIP AND CAMBRIDGE INTERNATIONAL SCHOLARSHIP (DECLINED)
· Granted full scholarship awarded to ~1% of applicants to pursue graduate study
- GORDON RESEARCH SEMINAR AND CONFERENCE ON GEOCHEMISTRY OF MINERAL DEPOSITS
· Granted \$300 as invited Gordon Research Seminar speaker, \$500 for the Gordon Research Conference
- 2021-2022 EURETTA J. KELLETT FELLOWSHIP
· Granted fellowship given academic distinction and potential for community contribution at the University of Cambridge; awarded full tuition and maintenance (£45,120) to pursue one year of graduate study
- 2022 EMMANUEL COLLEGE, UNIVERSITY OF CAMBRIDGE — POZZI FUND
· Examined intersection of science and art, sonified thermodynamic data from magma crystallization (£250)
- 2020 COLUMBIA UNIVERSITY — PHI BETA KAPPA, DEPARTMENTAL HONORS

RESEARCH EMPLOYMENT

- 2022-2024 DATA SCIENCE FELLOW IN GEOINFORMATICS, LAMONT-DOHERTY EARTH OBSERVATORY
Data Science Fellow Research
· Develop [mineralML](#), an open-source Python applying sequential, transfer learning machine learning (ML) solutions for probabilistic mineral classification and visualization
· Spearhead integration of data repository APIs of mineral, melt, and astromaterial data with Python
- 2021-2022 UNIVERSITY OF CAMBRIDGE
Master of Philosophy Research
· Develop and deploy ML solutions for electron microscopy data ($\sim 10^7$ pixels $\times$ 10) to automate mineral segmentation, reducing manual efforts (~ 10 s to ~ 5 hours/sample); create Bayesian algorithms for determining mineral chemistries
· Model plagioclase chemical profiles with finite-elements diffusion modeling to determine magma mixing timescales
- LAMONT-DOHERTY EARTH OBSERVATORY OF COLUMBIA UNIVERSITY
- 2019-2021 **Computational Research Assistant** (5/2020-7/2021); **Senior Thesis Research** (3/2019-5/2020)
· Developed [PyIRoGlass](#), an open-source Bayesian MCMC Python package to fit baselines to FTIR spectra for volatile phase (CO₂, H₂O) measurements within basaltic-andesitic glasses, creating the first standardized data processing method
· Identified the eruption trigger of fresh magma injection, occurring two weeks prior to outset, for the Volcán de Fuego eruption of 2018 by timing magma mixing with finite-element diffusion modeling
· Created novel arc melt thermometer with inverse theory to assess melt temperatures and uncertainties, improving accuracy and precision of reconstructed temperature
- 2018-2019 **NSF REU Summer Intern** (6/2018-9/2018); Independent Research (9/2018-1/2019)
· Analyzed and modeled *n*-alkane distributions and concentrations with dimensionality reduction techniques to reveal resolving power among plant functional types, in application to reconstructing hominid ecosystems

PAPERS

- 2026 [7, in review] **Shi, S.C.**, Wieser, P.E., Gordon, C., Toth, N., Antoshechkina, P., Gleeson, M., Lehnert, K., mineralML: Leveraging Machine Learning for Probabilistic Mineral Classification. *Submitted to Geochemistry, Geophysics, and Geosystems, preprint on Earth ArXiv*.
- [6] Tweedy, R., **Shi, S.C.**, Uno, K.T., Supervised Machine Learning Approaches to African Plant Chemotaxonomy from *n*-Alkane Chain Length Abundances. *Organic Geochemistry*, 105295.
- 2025 [5] Toth, N.*, **Shi, S.C.***, MacLennan, J., Tung, P.Y. (2025). EDS Analysis for Petrology: A Probabilistic Framework with GPyEDS. *JGR: Machine Learning and Computation*, 2(4), e2025JH000751. [* demarcates co-lead authors]
- [4] Wieser, P.E., **Shi, S.C.**, Gleeson, M., Rangel, B., DeVitre, C., Bearden, A., Lynn, K., Trusdell, F., Camille-Caumon, M. (2025). Fluid inclusion constraints on the geometry of the magmatic plumbing system beneath Mauna Loa: Part 1: Lavas and Tephros. *Bulletin of Volcanology*, 87(10), 1-24.
- [3] Gleeson, M., Wieser, P., DeVitre, C., **Shi, S.C.**, Millet, M.-A., Muir, D., Stock, M., Lissenberg, J. (2025). Persistent high-pressure storage beneath a near-ridge ocean island volcano (Isla Floraena, Galapagos). *Journal of Petrology*, 66(5), egaf031.
- 2024 [2] Moussallam, Y., Towbin, W.H., Plank, T.A., Bureau, H., Khodja, H., Guan, Y., Ma, C., Baker, M., Stolper, E.M., Naab, F.U., Monteleone, B.D., Gaetani, G.A., Shimizu, K., Ushikubo, T., Lee, H., Ding, S., **Shi, S.C.**, Rose-Koga, E.F. (2024). ND70 Series Basaltic Glass Reference Materials for Volatile Element (H₂O, CO₂, S, Cl, F) Analyses and the C Ionization Efficiency Suppressing Effect of Water in Silicate Glasses. *Geostandards and Geoanalytical Research*, 48(3), 637-660.
- [1] **Shi, S.C.**, Towbin, W.H., Plank, T.A., Barth, A.C., Rasmussen, D., Moussallam, Y., Lee, H., Menke, W. (2024). **PyIRoGlass**: An open-source, Bayesian MCMC algorithm for fitting baselines to FTIR spectra of basaltic-andesitic glasses. *Volcanica*, 7(2), 471-501.

CONFERENCES

- 2026 [26] Bawankule, N.N.A., Wieser, P., Downs, D.T., Gleeson, M.L.M., **Shi, S.C.**, Lynn, K.J., Bubble Talks and Crystal Clocks: Petrologic Insights into the 1790-1808 CE Kealaalea Hills eruption in Kilauea's Southwest Rift Zone (SWRZ) and Koa'e Fault System, AGU (*Poster*).
- [25] Antoshechkina, P., Deng, R., Lehnert, K., Nielsen, R., **Shi, S.C.**, Ustunisik, G.K., Towards a streamlined and sustainable curation pipeline for experimental petrology data, ExPet26 Conference (*Poster*).
- [24] Tweedy, R.R., **Shi, S.C.**, Uno, K.T., Leveraging Machine Learning Tools for Organic Geochemical Datasets: Opportunities and Pitfalls from an *n*-Alkane Chemotaxonomy Case Study, Gordon Conference on Organic Geochemistry (*Poster*).
- [23] **Shi, S.C.**, Wieser, P., Gleeson, M.L.M., Mutch, E.J.F., Lee, W.J.D., Plank, T., Lynn, K.J., Ruth, D., Mourey, A., Towbin, W.H., Martin, C., Moore, L.R., Rezeau, H., Kennedy, S.A., Tepley, F., Petrone, C.M., von der Handt, A., Probe the Probe, Test the Clock: A Round Robin Assessment of EPMA Variability and Olivine Diffusion Timescales, Goldschmidt (*Talk*).
- [22] **Shi, S.C.**, Wieser, P., Lee, H., Georgeais, G., Moussallam, Y., Sisson, T.W., Synthetic Reference Materials for Improved Volatile Quantification in Andesitic-Rhyolitic Silicate Melts: Application to the Sunset Amphitheater Eruption, Mount Rainier, Goldschmidt (*Poster*).
- [21] Wieser, P., **Shi, S.C.**, Turner, S., Gordon, C., Couperthwaite, F., Kent, A.J.R., Gazel, E., DeVitre, C., Ruth, D., Melt Inclusion and mineralogical constraints on mantle heterogeneity, pre-eruptive storage, and eruption initiation timescales for the 1 ka Paint Pot Crater Eruption, Medicine Lake Volcano, Goldschmidt (*Talk*).
- [20] **Shi, S.C.**, Wieser, P., Gleeson, M.L.M., Mutch, E.J.F., Lee, W.J.D., Plank, T., Lynn, K.J., Ruth, D., Mourey, A., Towbin, W.H., Martin, C., Moore, L.R., Rezeau, H., Kennedy, S.A., Tepley, F., Petrone, C.M., von der Handt, A., Probe the Probe, Test the Clock: A Round Robin Assessment of EPMA Variability and Olivine Diffusion Timescales, VolCali (*Talk*).
- [19] **Shi, S.C.**, Wieser, P., Gordon, C., Toth, N., Antoshechkina, P., Gleeson, M.L.M., Lehnert, K., mineralML: Leveraging Mineral Learning for Probabilistic Mineral Classification in Ore Systems, Gordon Conference on the Geochemistry of Mineral Deposits (*Poster*).
- [18] **Shi, S.C.**, Minerals, Models, and Uncertainty: An Open-Source Machine Learning Approach to Building Certainty in Ore Characterization, Responsible Raw Materials Conference: Creating Certainty in a World of Change (*Invited Talk*).
- 2025 [17] Wieser, P., **Shi, S.C.**, Gleeson, M., Bawankule, N., Ruth, D.C.S., Lynn, K.J., Downs, D., Martin, C., Kuehn, S.C., Moore, L., Energy Dispersive Spectroscopy (EDS) as a quantitative analytical tool for petrologic studies, GSA (*Talk*).
- [16] Moussallam, Y., Towbin, H., Plank, T., Bureau, H., Khodja, H., Guan, Y., Baker, M.B., Stolper, E., Naab, F., Monteleone, B.D., Gaetani, G., Shimizu, K., Lee, H., Ushikubo, T., Ding, S., **Shi, S.C.**, Rose-Koga, E.F., Development of Basaltic Glass Reference Materials for Volatile Element Analysis (H₂O, CO₂, S, Cl, F) and Investigation of Water-Induced C Ionization Suppression in Silicate Glasses Using SIMS, Goldschmidt (*Poster*).
- 2024 [15] Gleeson, M., Wieser, P., DeVitre, C., **Shi, S.C.**, Millet, M.-A., Muir, D., Stock, M., Lissenberg, J., Persistent Magma Storage in the Mantle Across 2.5 Myrs of Ocean-Island Volcanism, GSA (*Talk*).
- [14] **Shi, S.C.**, Wieser, P., Toth, N., Antoshechkina, P., Lehnert, K., mineralML: Leveraging Machine Learning for Probabilistic Mineral Classification, Gordon Seminar, Geochemistry of Mineral Deposits (*Invited Talk*).
- [13] Tweedy, R., **Shi, S.C.**, Uno, K.T., Machine Learning Analysis of *n*-Alkanes from Woody and Grassy African Plants, NE Geobiology Conference (*Talk*).
- 2023 [12] **Shi, S.C.**, Wieser, P., Toth, N., Antoshechkina, P., Lehnert, K., MIN-ML: Leveraging Machine Learning for Probabilistic Mineral Classification in Geochemical Databases, AGU (*Talk*).
- [11] Tweedy, R., **Shi, S.C.**, Uno, K.T., African Plant Functional Type Identification from *n*-Alkanes Chain Lengths via Non-Linear Methods, AGU (*Talk*).
- [10] Bidgood, A., **Shi, S.C.**, Prabhu, A., Que, X., Twigg, H., Nulff, M., Using Supervised and Unsupervised Machine Learning Methods to Predict Missing Geochemical Data and Determine Geochemical Trends in Multielement Systems: Application to Sediment-Hosted Ore Deposits, AGU (*Poster*).

	[9] Prabhu, A., Wong, M.L., Morrison, S.M.M., Ostroverkhova, A., Clark, M., Zhong, H., Prestgard, T.J., Li, W., Williams, J.R., Shi, S.C. , Mays, J., Hazen, R., From detecting agnostic biosignatures to characterizing chondrites: How network science is perfect for making scientific discoveries with geochemical data, AGU (<i>Invited Talk</i>).
	[8] Shi, S.C. , Wieser, P., Toth, N., Antoshechkina, P., Lehnert, K., MIN-ML: A Machine Learning Framework for Exploring Mineral Relations and Classifying Common Igneous Minerals, Goldschmidt (<i>Invited Workshop Talk</i>).
	[7] Shi, S.C. , Wieser, P., Lehnert, K., Profeta, L., MIN-ML: A Machine Learning Framework for Exploring Mineral Relations and Classifying Common Igneous Minerals, EGU (<i>Talk</i>).
2022	[6] Tweedy, R., Shi, S.C. , Uno, K.T., Grass in the Past: Eastern African Chemotaxonomy from Plant Wax <i>n</i> -alkanes, AGU (<i>Poster</i>).
	[5] Shi, S.C. , Barth, A.C., Plank, T.A., Towbin, W.H., Flores, O., Arias, C.P., Magma stalling weakens eruption: Uncertainty quantification in thermometry and volatile measurements, VMSSG (<i>Talk</i>).
	[4] Toth, N., Shi, S.C. , MacLennan, J., Automated petrography using machine learning, VMSSG (<i>Poster</i>).
2021	[3] Shi, S.C. , Barth, A.C., Plank, T.A., Towbin, W.H., Magma stalling weakens eruption, AGU (<i>Talk and ePoster</i>).
2018	[2] Shi, S.C. , Cerling, T.E., Uno, K.T., What plant is that? Chemotaxonomy from <i>n</i> -alkane molecular distributions of East African plants with implications for paleoecology, AGU (<i>Poster</i>).
	[1] Shi, S.C. , Cerling, T.E., Uno, K.T., Resolving taxonomy with <i>n</i> -alkane molecular distributions of East African plants, Columbia University Chandler Society Research Symposium (<i>Invited Talk</i>).

PROFESSIONAL ACTIVITIES

2025-2026	Goldschmidt Conference 2026 , <i>Session Co-Convener, From source to surface: advancing studies of magmatic systems with data-driven computational approaches</i>
2026	VolCali Conference 2026 , <i>Social Chair</i>
2024-2026	Gordon Conference on the Geochemistry of Mineral Deposits 2026 , <i>Gordon Seminar Co-Chair</i>

TEACHING AND MENTORING

	UNIVERSITY OF CALIFORNIA, BERKELEY
2025-2026	STEM*FYI Mentor Mentor STEM graduate students at Berkeley to increase success and retention.
2025-	Undergraduate Research Mentor Mentor undergraduate student, Natasha Bawankule, for project on diffusion chronometry at Kilauea.
2025	Graduate Student Instructor for EPS100B: Genesis and Interpretation of Rocks Teaching Evaluations: · Sarah never gave you the answer right away, she asked thought provoking questions that were engaging and facilitated deep thought. She was an excellent GSI! She also helped us out while also not giving away the answers, making each moment a learning moment. I felt that I actually learned in this course rather than having the answers fed to me.
	COLUMBIA UNIVERSITY
2023-2024	Invited Speaker at Goldschmidt Workshop (Open Data in Geochemistry: Navigating Present Data Infrastructure); German Mineralogical Society Data Science in Geochemistry and Cosmochemistry · Develop Jupyter Notebook on applying supervised/unsupervised machine learning for classification of large mineral datasets Guest Lecturer for Earth's Environmental Systems: Solid Earth · Present lecture on geochemical data, develop Jupyter Notebook for understanding petrologic trends in MORB
	UNIVERSITY OF CAMBRIDGE
2021-2022	Practical Demonstrator for Earth Sciences B (Second Year Module)
2021	Field Demonstrator for Cornwall Field Geology Trip (6 Days)

SERVICE

	UNIVERSITY OF CAMBRIDGE, DEPARTMENT OF EARTH SCIENCE
2021-2022	EDI, LGBTQ+ Working Groups , <i>Committee Member</i> · Perform outreach regarding equity, diversity, and inclusion
	COLUMBIA UNIVERSITY
2018-2020	Department of Earth Science Undergraduate Student Committee , <i>Founder and Co-Chair</i> · Developed seminar series to highlight climate and hazard research; initiated undergraduate involvement in equity outreach; spearheaded initiative to confer outstanding teaching awards to professors; led field geology trips alongside faculty
2016-2019	Peer Health Exchange , <i>Co-Coordinator and Diversity, Equity, and Inclusion Coordinator</i> · Taught classes regarding physical and mental health/wellness to NYC high schoolers; recruited the most diverse chapter of 150 educators in organization history; developed and presented equity trainings; managed \$50k budget

DATATHONS, WORKSHOPS, FIELD WORK

2026	Medicine Lake
2025	Stratigraphy and Earth History – Death Valley; Geologic Field Studies – Southwest U.S.; Geomorphology – South Lake Tahoe
2024	AVERT Project Field Workshop – Poás Volcano, Costa Rica
2023	Mineral Informatics Datathon – Earth and Planets Laboratory, Carnegie Institute for Science, <i>Attendee</i> Iceland Fieldwork – Cambridge Volcanology Group; Kenya Fieldwork – Uno Laboratory Goldschmidt Conference Workshop – Open Data in Geochemistry: Navigating Present Data Infrastructure, <i>Presenter</i>
	Mineral Informatics Datathon – University of Idaho, <i>Attendee</i>
2020	Goldschmidt Conference Workshop – Diffusion Chronometry, <i>Attendee</i>

ADDITIONAL INFORMATION

Languages	English (native), Chinese (native), French (working proficiency)
Computation	Python (PyTorch, TensorFlow), MATLAB, R, HTML/CSS, Bash/Unix, GitHub
Interests	Cycling, data sonification, music (production, journalism), fermentation, rowing